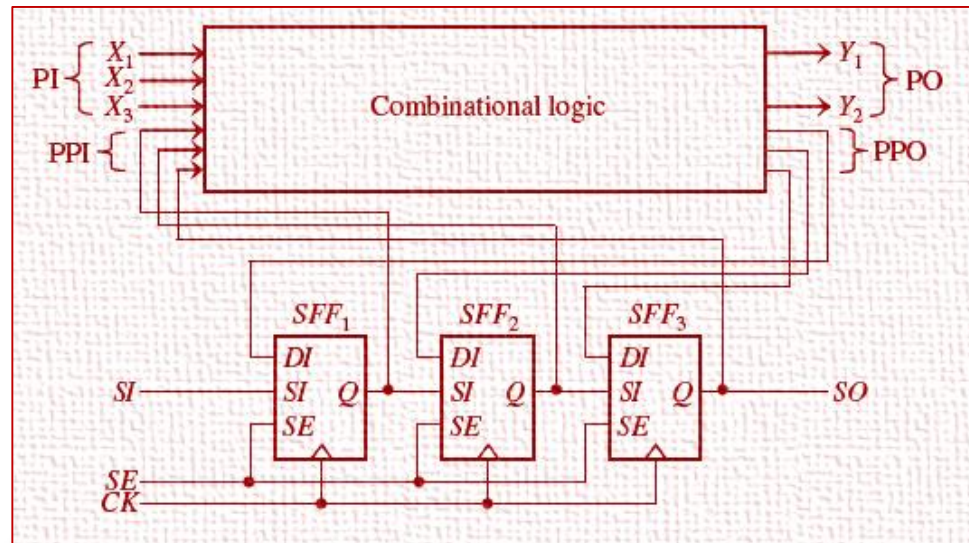


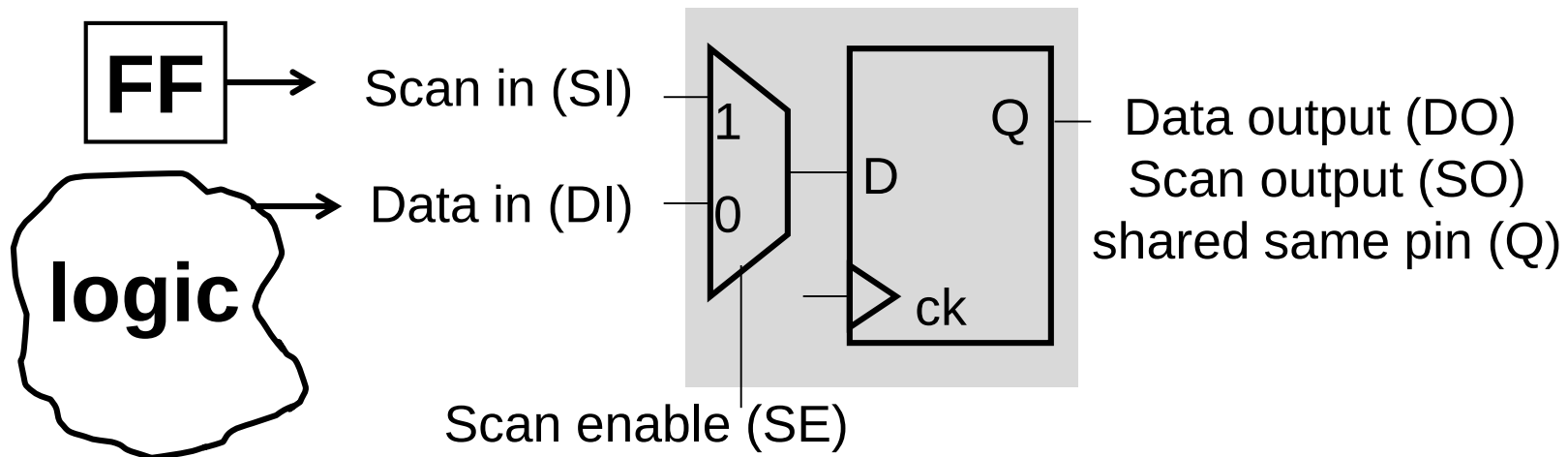
DFT - Part 1

- Introduction
- Internal Scan
 - ◆ FF-based
 - * MUXed-D scan (1973, Stanford)
 - MUXed-D scan flip-flop
 - Test Mode Operation
 - * Clocked scan (1968, NEC)
 - * Other scan
 - ◆ Latch-based
 - * LSSD (1977, IBM)
- Scan Design Flow
- Issues and Solutions
- Conclusion



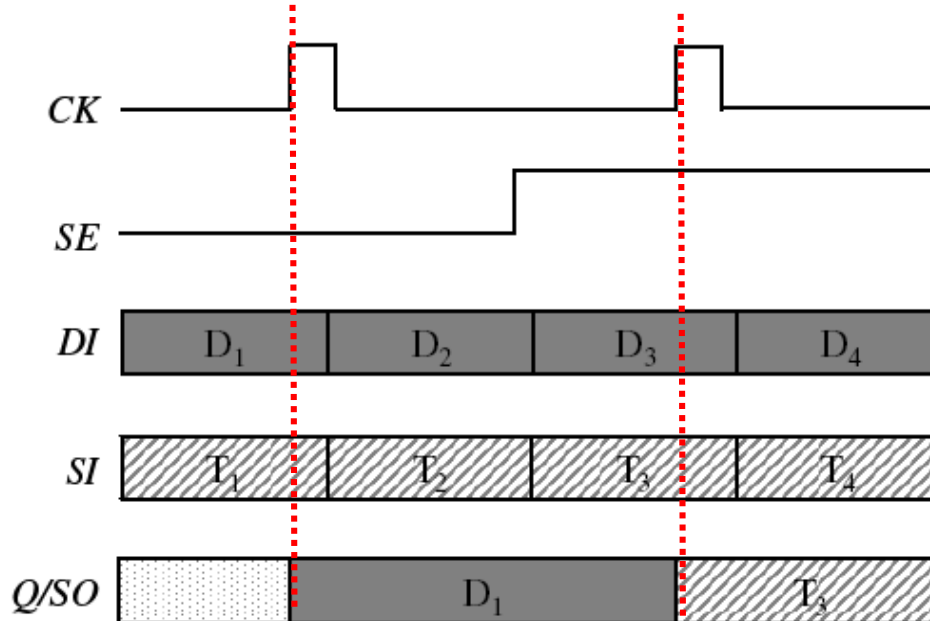
MUXed-D Scan Flip-flop [Williams 73]

- Often used in flip-flop based, standard cell design
- 3 input, 1 output, 1 clock pin
- **Scan Enable (SE)** pin selects inputs
 - ♦ **Data in (DI)**: from logic
 - ♦ **Scan in (SI)**: from previous scan FF
- **Data output (DO)** and **Scan output (SO)** often share same pin (Q)
- One multiplier (MUX) slower and larger than regular non-scan FF

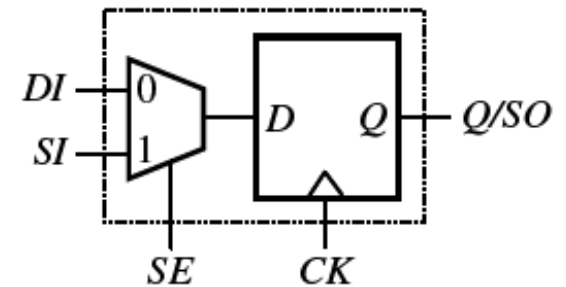


MUXed-D Scan Flip-flop

- Positive edge triggered MUXed-D scan FF (WWW Fig. 2.9)
 - ♦ **SE=1**, SI → Q at positive clock edge
 - ♦ **SE=0**, DI → Q at positive clock edge



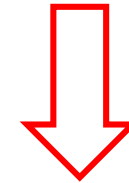
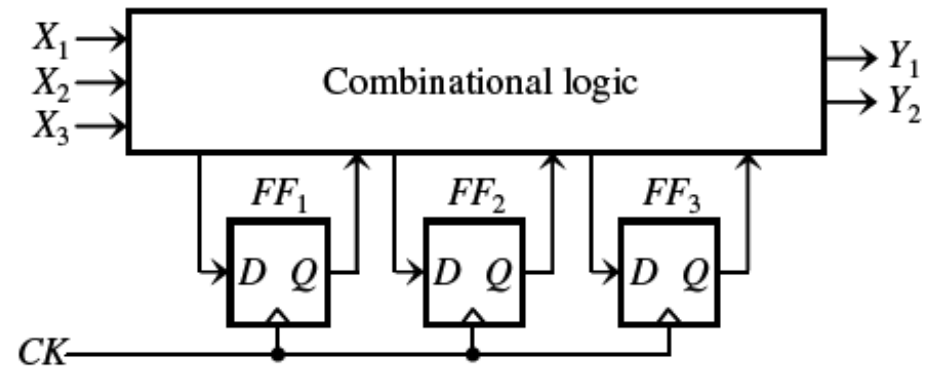
	operation
SE=0	Capture (DI→Q)
SE=1	Shift (SI→SO)



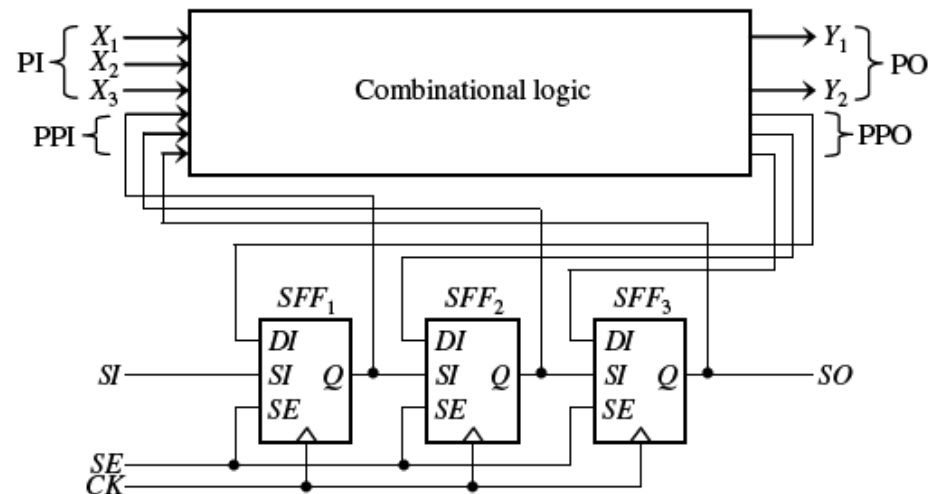
SE = 1 enables scan

Muxed-D Scan Architecture

- Original circuit
 - ♦ Single clock
 - ♦ Positive edge triggered
 - ♦ Non-scan DFF



- After scan insertion
 - ♦ Three extra pins
 - * SI, SE, SO

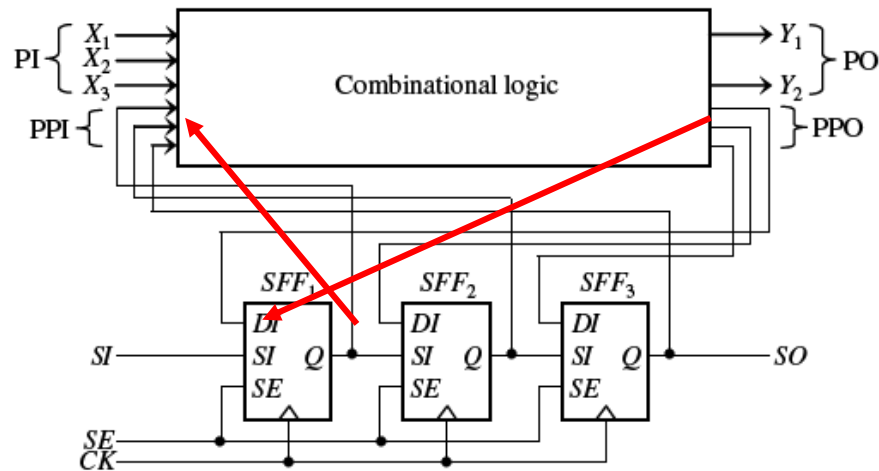


SFF = scan FF

Normal Mode Operation

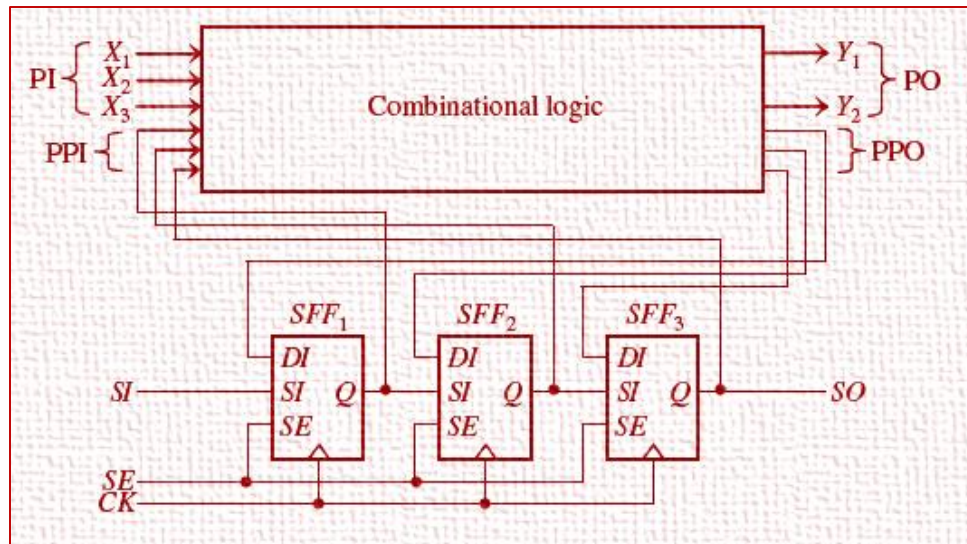
- Always $SE = 0$
- Same as original circuit

	operation
SE=0	Capture ($DI \rightarrow Q$)
SE=1	Shift ($SI \rightarrow SO$)

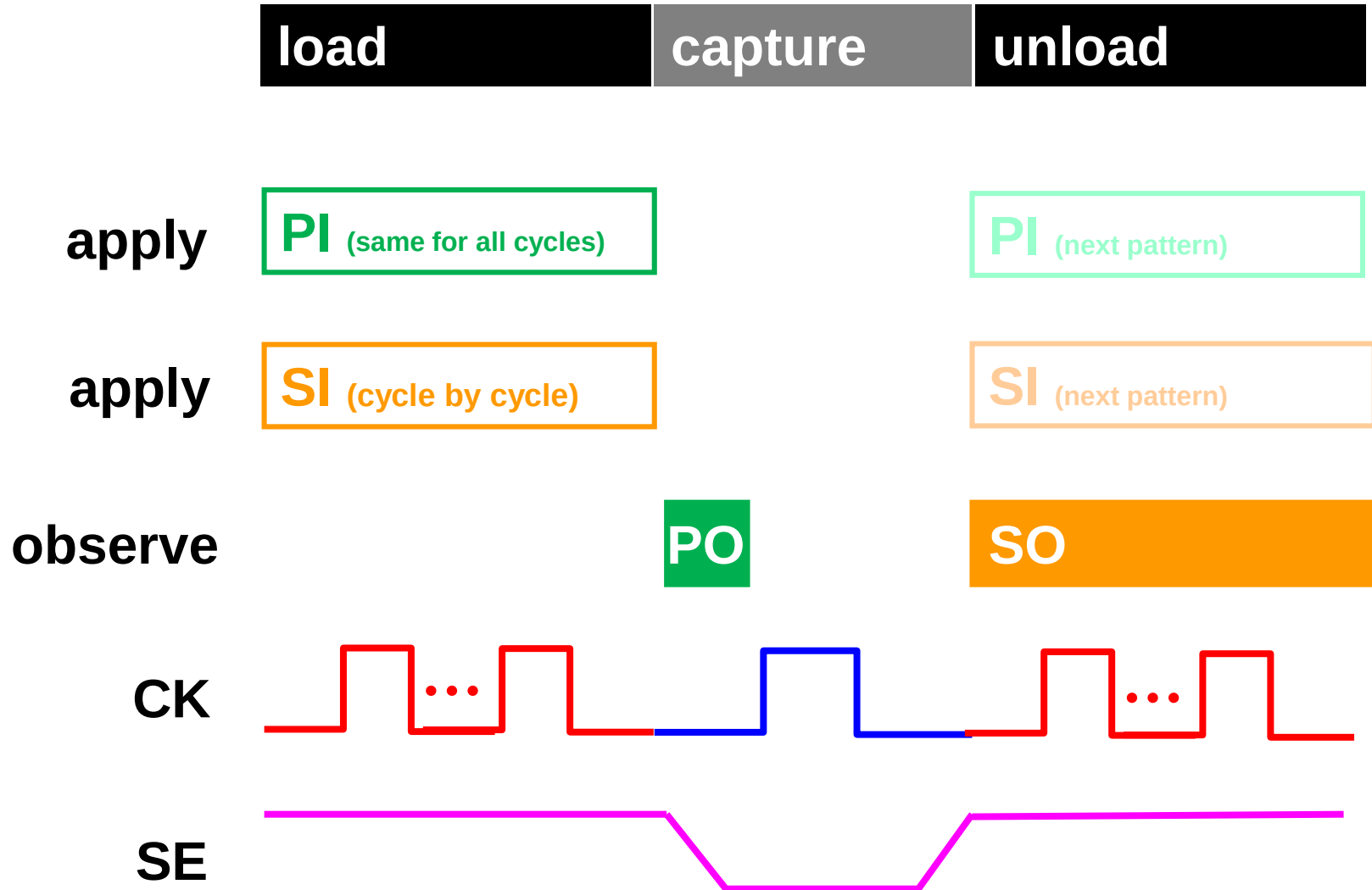


Test Mode Operation

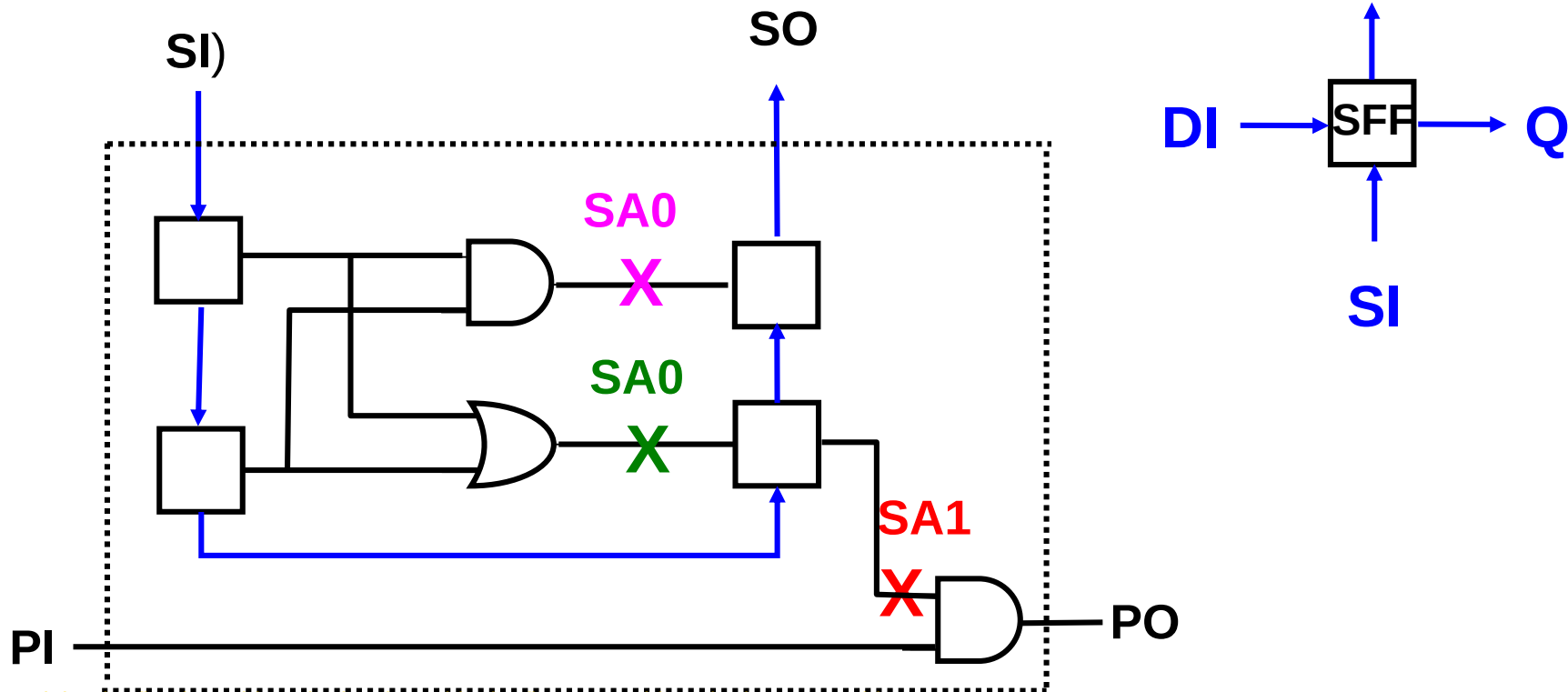
- Stuck-at fault testing
 - ◆ One pattern per scan
- Delay fault testing
 - ◆ Two-patterns per scan
 - * Launch-on-Shift
 - * Lunch-on-Capture



SAF Operation



SAF Example



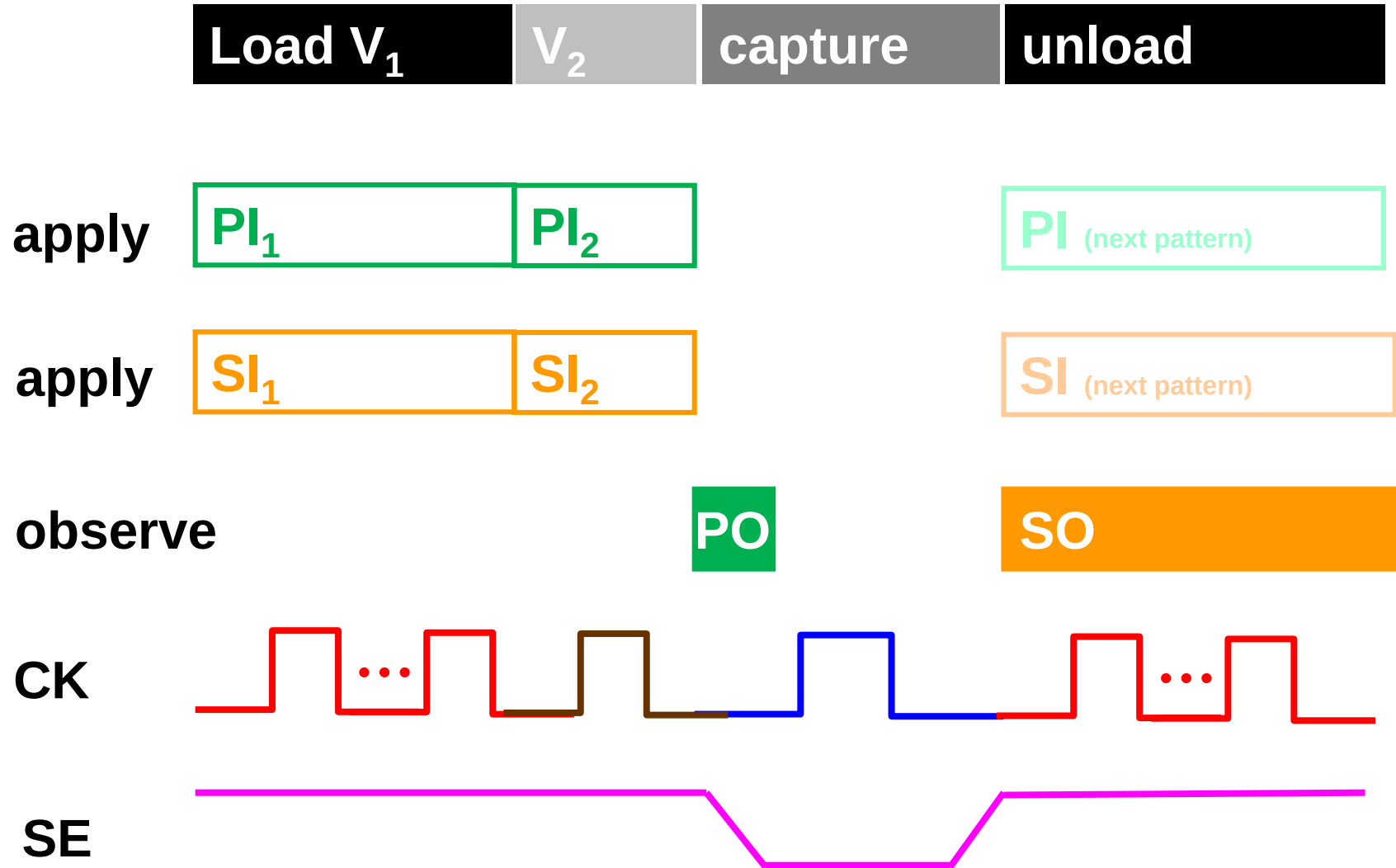
load		capture		unload
CK = PPPP		CK=P		CK = PPPP
SE = 1111	SE 1→0	SE = 0	SE 0→1	SE = 1111
SI = X011				SO = H H XX
PI = 1	PO = L			

P = CK pulse
H=expected 1
L=expected 0
Left bit first in/out

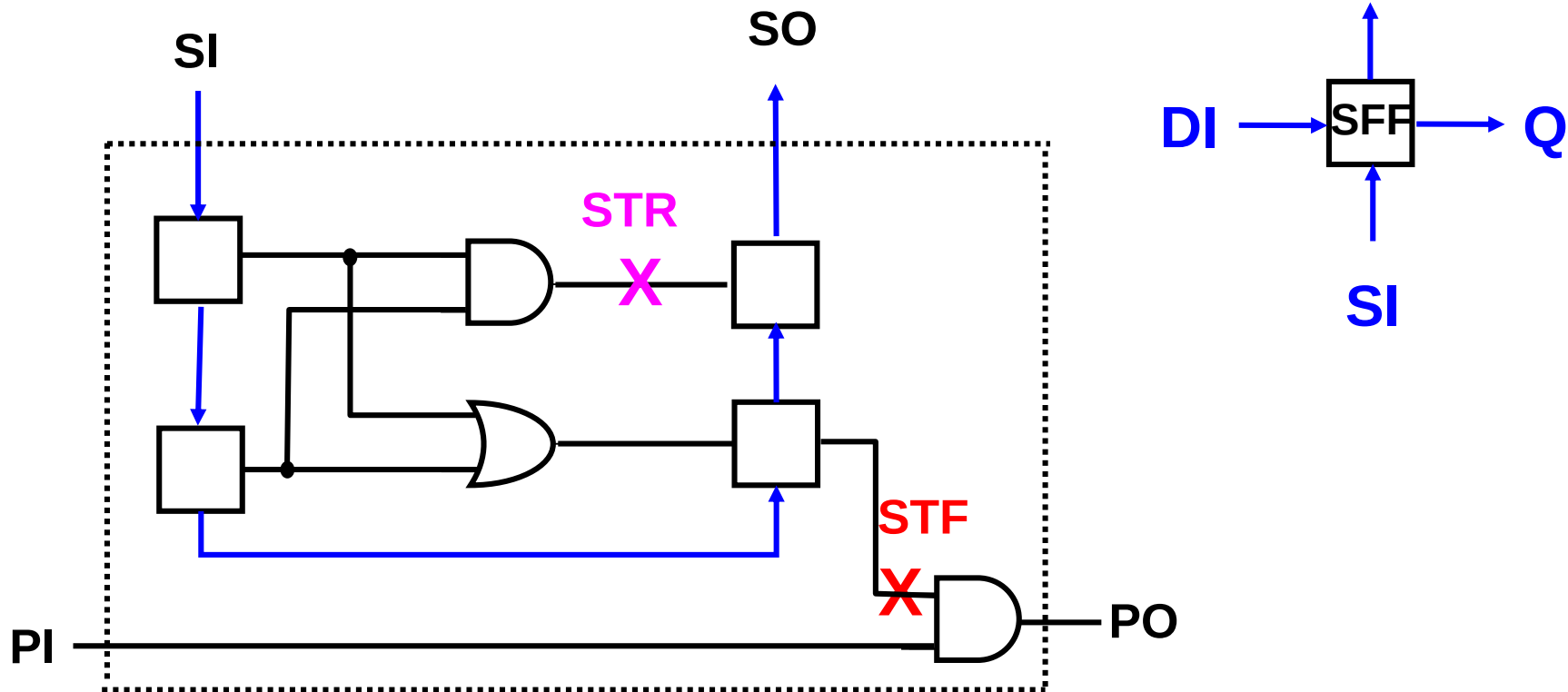
Test Mode Operation

- Stuck-at fault testing
 - ◆ One pattern per scan in/out
- Delay fault testing
 - ◆ Two-patterns per scan in/out
 - * Launch-on-Shift (aka. Skew load)
 - * Launch-on-Capture (aka. Broad-side load)

LOS Operation



LOS Example



load V_1	V_2		Capture		unload
CK = PPPP	CK=P		CK=P		CK = PPPP
SE = 1111	SE=1	SE 1→0	SE = 0	SE 0→1	SE = 1111
SI = X101	SI=1				SO = HXXX
PI = X	PI=1	PO=L			

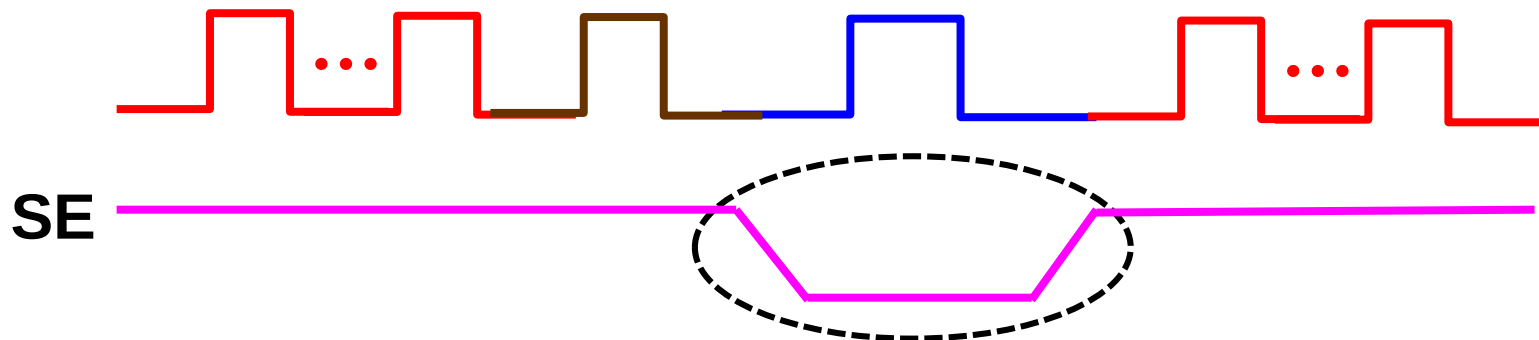
LOS -- Pro and Cons

- Advantages

- + Easier for combinational ATPG

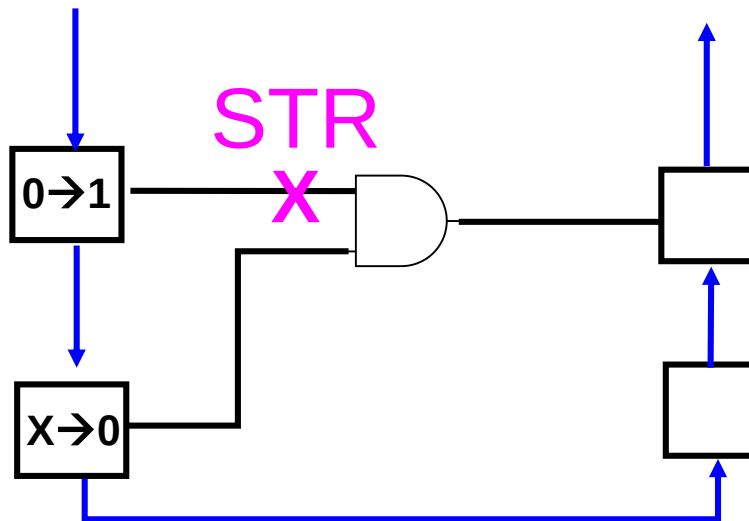
- Disadvantages

- V_2 -capture time is not at-speed
 - » Scan_enable signal fall time limitation
 - » Currently, ~ 5ns minimum
 - » Solution: route scan_enable as clock (area overhead)
 - Invoke transitions that never happen in normal operation
 - Some TDF not testable; **structure dependency problem**
 - » V_2 is shifted version of V_1



Structure Dependency

- STR fault untestable using LOS
- This fault can actually happen in normal operation
 - ♦ This fault needs to be tested, but how?



Some faults are untestable by LOS

Test Mode Operation

- Stuck-at fault testing
 - ◆ One pattern per scan in/out
- Delay fault testing
 - ◆ Two-patterns per scan in/out
 - * Launch-on-Shift (aka. Skew load)
 - * Lunch-on-Capture (aka. Broad-side load)

LOC Operation



apply



apply

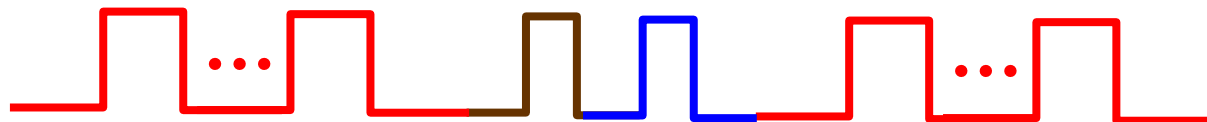


observe

P
O



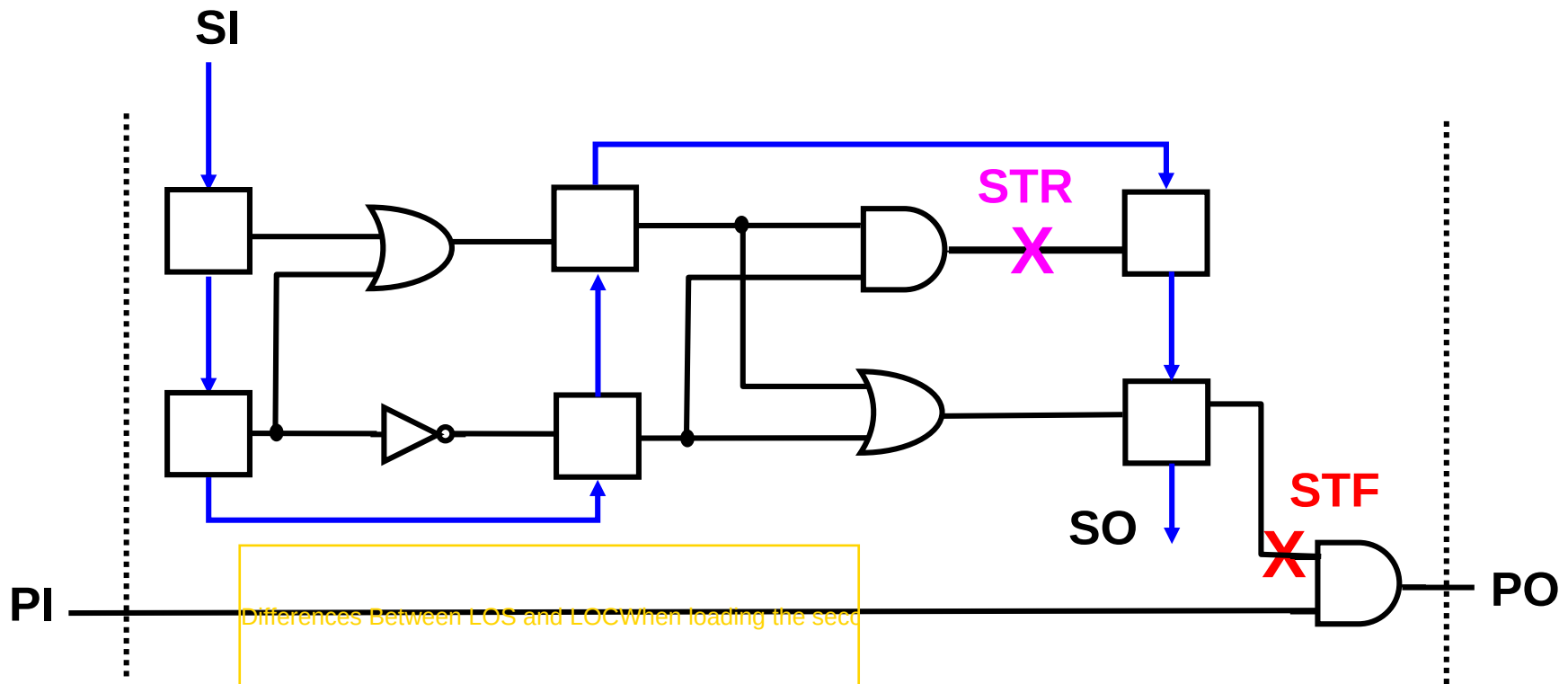
CK



SE



LOC Example



Load V_1		V_2		capture		unload
CK=PPPPPP		CK=P		CK=P		CK=PPPPPP
SI=1X0001		SI=X				SO=X H XXXX
SE=111111	SE 1→0	SE=0	SE=0	SE=0	SE 0→1	SE=111111
PI=X	PI=1		PO=L		SO=X	

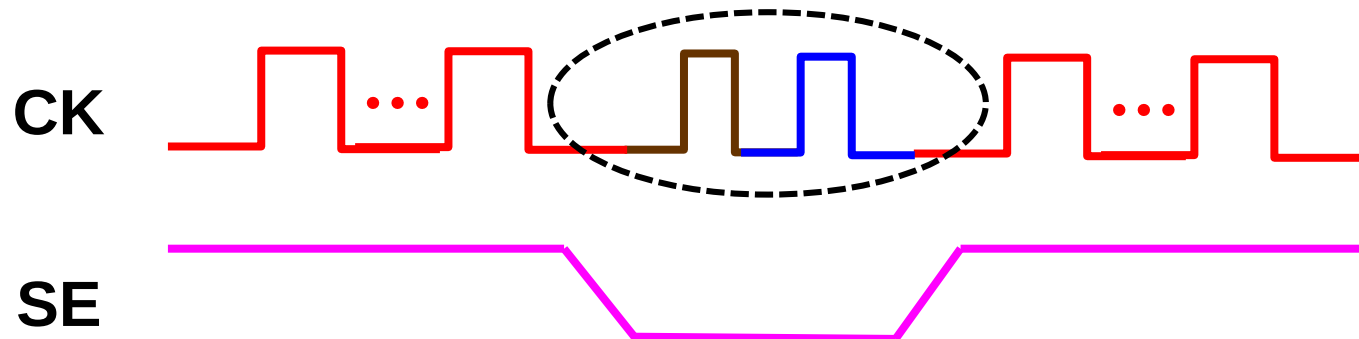
LOC -- Pro and Cons

- Advantages

- + V_2 -capture can be applied at system clock speed
 - + True at-speed testing
- + Avoid structure dependency problem of LOS
- + Minimum work in physical design, SE can be routed as normal signal

- Disadvantages

- ATPG is more complex
 - » Requires two time frames



FFT

- **Q: Are there faults untestable by LOC?**
 - ♦ **If so, how should we test them?**